Chemical Engineering Laboratory I

CHE 381

Welcome!!

UNIVERSITY OF ILLINOIS AT CHICAGO

Today's Schedule

Today we will:

- Go over syllabus & lab schedule
- Discuss safety in the laboratory
  - Read and sign safety document
  - Safety Assignment I
- Assign lab teams and first lab
- Move to lab and get familiar with lab/experiments

Chemical Engineering Laboratory I
Room 120 CEB

Morning: Thursday 9:00AM – 12:30 PM

Afternoon: Thursday 1:00PM – 4:30PM

Instructor: Alan Zdunek

- Phone: 312-996-4607
- E-mail: zdunek@uic.edu

Teaching Assistants: **Tiep Pham**

- Email : tpham37@uic.edu

Walt Schindler, Laboratory Shop Supervisor Rm 125

John Sitasz, Computer/Electronics Engineer Rm 107

Syllabus

Course Objectives

- Experimental verification of theoretical principles
- Familiarization with industrial processes and equipment

Development of:

- ***Hands-on experience*** and an ***engineering sense***
- ***Teamwork skills***: participation and leadership
- ***Technical communications*** (oral and written)
- ***"Time value";*** project planning, implementation and completion.

Student Learning Objectives

- **Integrate** knowledge and skills from prior chemical engineering courses.
- **Design and conduct** realistic experiments, analyze data, draw concrete conclusions and make relevant recommendations.
- **Solve** complex, open-ended problems creatively, in a team setting.
- **Communicate** effort and findings effectively through the written and oral word.

Initial Safety Document

To be read, signed and handed in today

1. You must wear standard safety glasses in the lab at all times. You must wear chemical workers' goggles for any operation with hazardous chemicals.
2. Adequate leg protection is required (no shorts).
3. Full foot protection is required (no open-toed shoes; try boots if possible).
4. Use disposable nitrile gloves for hand protection when handling any chemical.
5. If you wear contact lenses, you must wear chemical workers' goggles in the lab at all times.
6. In the event of a fire or other accident, contact your TA immediately. Please review the location of the laboratory fire extinguishers. Know how to call emergency services.
7. Cuts or burns must be treated immediately at the student health center. Transportation will be arranged for you.
8. Neither food nor drink is allowed in the laboratory.
9. Please use rubber suction to fill pipettes.
10. Secure long hair, ties(?), or loose jewelry. Long unrestrained items can become entangled in moving equipment. This could damage equipment.
11. Unauthorized experiments are not allowed (Do that stuff at home). Never work in the laboratory alone. No experimental work outside of lab times is allowed unless authorized by ther instructor or TA.
12. Be sure every electrical device is plugged in and well grounded before you use it. When in doubt, please ask!

Guidelines on Safety Glasses

- **Safety glasses must always be worn on the first floor of the lab at all times.**
- Pick up a pair when you enter the laboratory in the morning and put them on.
- **Compliance will be monitored by the instructor and TA during the lab period.** Non-compliance will be recorded.
  - 3 instances of non-compliance will result in a zero on the Individual Assessment grade.

Other Safety Considerations

- High-pressure gas cylinder usage (Membrane Separation); ask TA or Instructor for guidance before using the compressed gas cylinder.
- Chemical usage (Carbopol, calcium carbonate)
  - Read MSDS sheets to familiarize yourself with hazards of each chemical
  - Wear proper safety attire when handling (gloves, dust mask)
- Know where the safety shower, eyewash and fire extinguishers, first aid kit, MSDS are located

CHE381 Experiments

A: Viscosity of Non-Newtonian Liquids

B: Evaporation

C: Fluid Flow in Pipe Systems

D: Fluidized Bed

E: Filtration

F: Membrane Separation

Primary Lab Information

- Lab manuals: Most up-to-date versions available through Blackboard

- **Unit Operations in Chemical Engineering textbook**
  - **Great resource for theory and practical information**
  - **USE IT!!**

Additional Resources

- Additional information is in Blackboard for each experiment
- Perry’s Chemical Engineering Handbook
  - Another great resource for practical info and correlations.
- Online resources/UIC library resources

- NOTE: At least 4 references are required for each final lab report (Not including lab manual)

Laboratory Schedule Fall 2015: Thursday

Week 1:	Aug	27	Introductions; lab familiarization	
Week 2:	Sep	1	**Safety Assignment I Due**	
	Sep	3	Pre-lab	**LAB 1**
Week 3:		10	Laboratory	
Week 4:		17	Pre-lab	**LAB 2**
Week 5:		24	Laboratory	
Week 6:	Oct	1	Pre-lab	**LAB 3**
Week 7:	Oct	8	Laboratory	
Week 8:		15	Pre-lab	**LAB 4**
Week 9:		22	Laboratory	
Week 10:		29	Pre-lab	**LAB 5**
Week 11:	Nov	5	Laboratory	
Week 12:		12	Pre-lab	**LAB 6**
		18	**Safety Assignment II Due**	
Week 13:		19	Laboratory	
Week 14:		**26**	**Thanksgiving Week (No Lab)**	
Week 15:	Dec	3	Class presentations (**Final Lab Period)**	
Week 16:		8-12	**FINAL EXAM WEEK (No Lab)**	

Course Output

- Pre-Lab Plans
- Final Lab reports
- Oral presentations (every two weeks)
- Safety Assignments:
  - Safety hygiene plan and quiz
  - SACHE tutorial
- Final oral presentation (end of semester)

Lab and Assignment Cycle

Two Lab periods for each experiment

- **Period 1 (Lab Prep)**
  - Familiarization with equipment
  - Lab tasks assigned by team
  - Perform initial measurements and calibrations
  - Complete Pre-Lab Plan with Safety Analysis
- Pre-Lab Plan due at end of lab period

Lab and Assignment Cycle

Period 2:

- Devoted to performing the experimental work
- Collecting data, analyzing results, creating graphs, figures and tables, error analysis.
- Write Final Lab Report
- Prepare oral presentation for next lab period

Final Lab Report due:

- The next Wednesday at 1PM

- **Next Period 1:**
  - Oral presentation to incoming team

NOTE on 1st Lab Schedule

1st pre-lab period will start next week, Sept 3rd:

<u>1st Pre-Lab Plan due:</u>

Thursday September 3th

End of lab section, Morning and Afternoon

<u>1st Final Lab period: September 10th</u>

1st final lab report due Wednesday Sept 16th

Pre-Lab Plan

- Briefly states the objective of the experiment, the experimental procedure to be followed and a safety analysis of the equipment. **(Length: 4 pages or less)**

Lab Prep Report Sections:

- Title page
- Lab Objective
- **Experimental**
- Safety Analysis Sheet

See PreLabPlanTemplate.doc for more details

Final Lab Report

- Tells what you did, why you did it, what you learned, and the significance of your results.
- **Length ≤10 pages** (not including Appendices (Raw Data, Error Analysis, Calculations, Individual Contributions))
- Key to writing an excellent report lies in presenting concisely your response to the objectives of the project and justifying that response with appropriate documentation.

Final Lab Report Sections

- Executive Summary (industrial-type format)
- Results
- Discussion
- Conclusion
- References (additional references if applicable)

Appendices

- Data Tabulation
- Error Analysis
- Individual Team Contributions

See FinalLabReportTemplateFall2014.doc for more details

Lab Reports: A Few Suggestions

- Keep report length within the guidelines!!
- Use SI units
- **Check unit consistency in equations**
- Clearly identify all variables (with units).
- Create easy-to-understand figures, graphs, tables
  Key to properly presenting your data;
  - Show X and Y axis titles and units clearly.
  - Use concise figure captions and titles.
  - Don’t forget to refer to and explain your figure/graph/table in the text!!

Acceptable graph and figure title (except for Y-axis; no units)

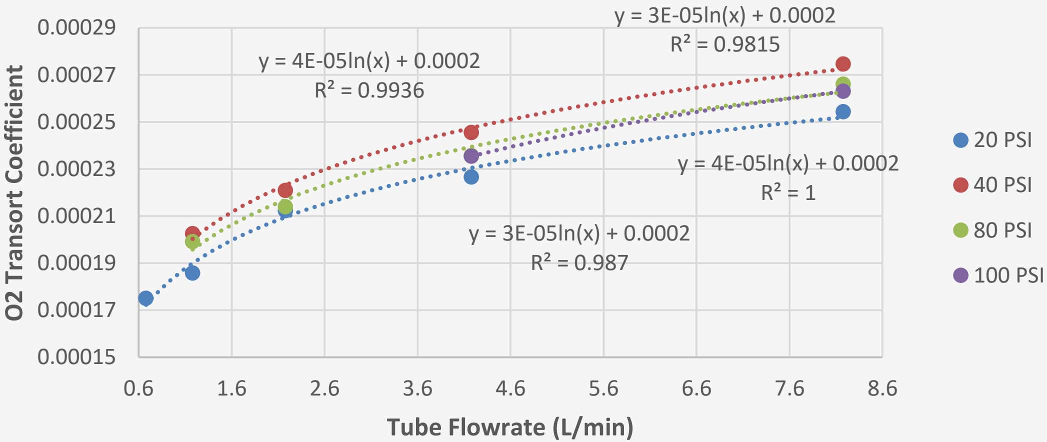

Figure 1: O_2 Transport Coefficients for Parallel Configuration

Acceptable Table
(except for significant figures)

Table 1: Raw data collected from Parallel experiment

Run	Pressure (psig)	Tube Flow (m^3/s)	Shell Flow (m^3/s)	O_2 Level Tube	O_2 Level Shell
1	20	1.13333E-05	1.03333E-05	0.138	0.273
2	20	1.96667E-05	1.86667E-05	0.147	0.285
3	20	3.63333E-05	3.53333E-05	0.159	0.304
4	20	6.96667E-05	6.86667E-05	0.177	0.325
5	20	0.000136333	0.000135333	0.189	0.343
6	40	1.96667E-05	1.86667E-05	0.096	0.287
7	40	3.63333E-05	3.53333E-05	0.108	0.308
8	40	6.96667E-05	6.86667E-05	0.126	0.337
9	40	0.000136333	0.000135333	0.149	0.37
10	80	1.96667E-05	1.86667E-05	0.058	0.279
11	80	3.63333E-05	3.53333E-05	0.064	0.294
12	80	6.96667E-05	6.86667E-05	0.077	0.323
13	80	0.000136333	0.000135333	0.098	0.363
14	100	6.96667E-05	6.86667E-05	0.062	0.317
15	100	0.000136333	0.000135333	0.083	0.356

Lab Report Suggestions

- **Results/Discussion section**: provide a specific, detailed discussion when presenting the analysis of your data that supports your conclusions.
- References listed in Reference Section ***must be cited*** in the body of the report.
  - References are not only for the Theory section: You can also cite a reference in other sections (such as the Results or Discussion section if needed to explain a particular result or to compare your result with literature values.
- Don’t start a heading at the bottom of the page (ignore the 2-space rule when this occurs)

Oral Presentations

- After completing an experiment, the team will make a 10-15 minute presentation to the next incoming group (pre-lab period).
- Brief but concise presentation describing:
  - Objective
  - Experimental procedures
  - Experimental challenges, safety
  - Lessons learned
- Use presentation software (see template in BB)
- Make copy and give to TA for grading

Project Teams

- 5-6 students per team
-
- Team assigns a team leader
- Team members rotate responsibilities for different experiments (everyone takes a turn)
- Instructor and teaching assistant reserve right to move students to different teams

Attendance

- Attendance is required for all lab periods.
  - TA will take attendance; please check in before lab starts.
    - On-time: $\leq$ 15 minutes
    - Late: $\geq$15 minutes to $\leq$50 minutes
    - Absent: > 50 minutes
  - If you have to miss a lab you must personally contact instructor or TA Wednesday night. Email is best. Excused absences may require proof.
  - Attendance will be part of individual assessment grade.

Grade Determination

Pre-Lab Plan	20%
Final Lab Report	40%
Oral Presentations	10%
Safety quiz/ SACHE tutorial	10%
Individual Performance Assessment	20%
	100%

UIC Blackboard Site

- Blackboard website will be used for general class announcements, notes and documents, and any supplementary materials and information.
- Grades will also be posted on BB

Safety Assignment I:
Industrial Chemical Hygiene Plan

- A Chemical Hygiene Plan (CHP) and Safety Rules from a corporate R&D Center have been uploaded into the Course Documents-Safety folder. CHP's are a required safety document at R&D facilities.

Safety Assignment: Due Tues Sept 1st

- READING ASSIGNMENT: You will be responsible for material in the following Chapters: Chapter 1, 3.2, 4, 5, 6, 7, 8. and the Definitions at the End. *SKIP Chapter 2, 3.1 and 3.3 and the Appendices and Attachments (not included).*
- SAFETY QUIZ: Due Tuesday, September 1st at 5pm. (CHE 381 Safety Quiz.doc is located in the Safety folder). Be brief but concise in your answers. Open book/open notes and team discussion is allowed but you must fill out the quiz yourself.

Safety Assignment II: SACHE Tutorial, Chemical Reactivity

AICHE (American Institute of Chemical Engineers) has created a website (SACHE) with safety tutorials covering process safety, risk assessment and chemical reactivity hazards.

- ASSIGNMENT: Complete tutorial **"Chemical Reactivity Hazards"** in the website.
- GRADING: You must email proof (Certificate issued by SACHE) that you have completed the tutorial and answered the questions for full credit.
- **DUE: Wednesday November 18th at 5pm**

Lab Team Set Up